MADX MOMENTS

Capturing Moments, Creating Memories · Live Photography Automation

Project Proposal & Budget
Date: November 2026
Document: Final Infrastructure Plan

Cloud Infrastructure & Cost Estimation Report

Complete Monthly and Yearly Cost Calculation for 15 Events / Month (30,000 Photos)

> **★ Executive Summary & Key Finding**
>
> By hosting our entire web platform (**Frontend Next.js, Backend FastAPI, and AI Face Search PostgreSQL Database**) on a single dedicated **Hostinger KVM 2 VPS** combined with **Cloudflare R2 Object Storage** (zero egress fees), our total platform cost is **only ₹1,132 per month (₹13,587 per year)**. This achieves an ultra-low operational cost of **only ₹75.50 per event**.

1. Monthly Operational Baseline

Parameter	Monthly Volume	Yearly Volume (12 Months)	Operational Remarks
Total Events	15 Events / mo	180 Events / yr	Weddings, receptions, studio shoots
Photos per Event	2,000 Photos	—	Transferred via camera Wi-Fi / SD card
Total Ingested Photos	**30,000 Photos**	**3,60,000 Photos**	Watermarked & AI face-indexed
Active Event Guests	~3,000 – 5,000 Guests	~36,000 – 60,000 Guests	QR code scanning & selfie search

2. Cost Breakdown: Scenario A — High-Resolution JPEG RECOMMENDED (99% INDUSTRY STANDARD)

Camera outputs Full-Res 4K/8K JPEG (6 MB to 8 MB per photo). Pin-sharp for 12x18 inch album printing, 4K display, and instant mobile viewing. Storage consumed: ~180 GB / month.

Component	Service / Plan	Specifications	Monthly Cost	Yearly Cost
All-in-One Server (Frontend + Backend + DB)	Hostinger KVM 2 VPS	2 vCPU, 8 GB RAM, 100 GB NVMe, 8 TB Bandwidth (Runs Next.js + FastAPI + Postgres pgvector + Free SSL)	₹699	₹8,388
Photo Cloud Storage & CDN	Cloudflare R2	180 GB/mo @ $0.015/GB (First 10 GB Free). **Zero Egress/Download Fee**	₹350	₹4,200
Custom Domain & Branding	Domain Registrar	Annual renewal (e.g. madxmoments.com)	₹83	₹999
AI Face Recognition Engine	InsightFace (Self-Hosted)	Runs directly on VPS CPU. Zero external per-face API charges.	₹0	₹0
TOTAL (Scenario A - High-Res JPEG)			**₹1,132 / mo**	**₹13,587 / yr**

Average Operational Cost per Event: ₹1,132 ÷ 15 = ₹75.50 only!

3. Cost Breakdown: Scenario B — Uncompressed RAW Master Files RAW ARCHIVAL ONLY

Uploading uncompressed camera RAW files (Sony .ARW / 100 MB per photo) directly to the cloud for lifetime raw archive. Storage consumed: ~3,000 GB (3 TB) / month.

Component	Service / Plan	Specifications	Monthly Cost	Yearly Cost
All-in-One Server (Frontend + Backend + DB)	Hostinger KVM 2 VPS	2 vCPU, 8 GB RAM, 100 GB NVMe (Next.js + FastAPI + Postgres + Free SSL)	₹699	₹8,388
Photo Cloud Storage & CDN	Cloudflare R2	3,000 GB (3 TB) / month. **Zero Egress/Download Fee**	₹3,800	₹45,600
Custom Domain & Branding	Domain Registrar	Annual renewal (e.g. madxmoments.com)	₹83	₹999
AI Face Recognition Engine	InsightFace (Self-Hosted)	Runs on VPS CPU. Zero external per-face API charges.	₹0	₹0
TOTAL (Scenario B - RAW Archival)			**₹4,582 / mo**	**₹54,987 / yr**

Average Operational Cost per Event: ₹4,582 ÷ 15 = ₹305.50 per event.

4. Side-by-Side Comparison & Annual Savings

Comparison Metric	Scenario A (High-Res JPEG)	Scenario B (100 MB RAW)	Strategic Recommendation
Photo Storage Format	4K Full-Res JPEG (6–8 MB)	Sony ARW RAW (100 MB)	Scenario A is 10x faster to transmit over venue Wi-Fi
Monthly Cloud Storage	180 GB / month	3,000 GB (3 TB) / month	Scenario A saves 2.82 TB of bandwidth monthly
Total Monthly Cost	**₹1,132 / month**	**₹4,582 / month**	**Scenario A saves ₹3,450 every month**
Total Yearly Cost	**₹13,587 / year**	**₹54,987 / year**	**Net Annual Savings: ₹41,400**
Operational Cost per Event	**₹75.50 / event**	**₹305.50 / event**	Negligible expense against event package pricing

5. Technical Architecture on Hostinger VPS + Cloudflare R2

Hostinger KVM 2 VPS (₹699/mo) Hosts:

- **Frontend**: Next.js 15 (Node.js runtime) for client & guest portal
- **Backend**: Python FastAPI REST API with Uvicorn worker pool
- **Database**: PostgreSQL 16 with `pgvector` extension for AI embeddings
- **AI Face Engine**: InsightFace model running on local CPU (0 API fees)

Cloudflare R2 Object Storage (₹350/mo):

- Stores master photos, watermarked photos, and face thumbnails
- **100% Free Egress (Downloads)**: Zero bandwidth bill for guest downloads
- Global Edge CDN: 300+ worldwide locations for sub-second photo access
- Unlimited auto-scaling: Never runs out of disk space

- **Web Server**: Nginx reverse proxy with automated Let's Encrypt SSL

6. Field Camera Workflow & Zero Data Loss Guarantee

Zero-Install Laptop Setup: The sync agent (`start-sync.bat`) runs directly from a standard USB pen drive on any laptop. Cameraman enters the Event ID, and the camera sends photos wirelessly via Wi-Fi FTP.

- **Disconnections & Power Failures**: Photos remain 100% safe on the camera SD card. On reconnection, sync auto-resumes seamlessly without creating duplicate uploads.
- **Mobile Phone Option (No Laptop)**: Photographers can upload directly from their smartphone via an OTG SD card reader or camera Wi-Fi app in 2 minutes.

Prepared by: Engineering & Architecture Team

MADX Photography · Confidential & Proprietary